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Do not solve.

MA3521 — Tuesday/Wednesday Worksheet Key

Name:

1. ☑ Linear ☐ Separable $\qquad ty' + 2y = e^t \qquad y(1) = 0$

Show a suitable intermediate step

$$(t^2 y)' = te^t$$

Standard form $y' + (2/t)y = e^t/t$

$$y' + p(t)y = q(t), \quad p(t) = 2/t, \quad q(t) = e^t/t$$
$$\mu(t) = \exp\left(\int p(t)\,dt\right)$$
$$\int (2/t)\,dt = 2\ln|t|$$
$$\mu(t) = e^{2\ln|t|} = |t|^2 = t^2 \quad (t > 0 \text{ since } t_0 = 1)$$
$$\int te^t\,dt = e^t(t-1) + C$$
$$t^2 y = e^t(t-1) + C$$
$$y(1) = 0 \implies 0 = e(1-1) + C \implies C = 0$$

Show a fully integrated answer

$$t^2 y = e^t(t-1)$$

2. ☐ Linear ☑ Separable $\qquad y' = 2t\cos(t^2)(1+y^2) \qquad y(\sqrt{\pi}) = 0$

Show a suitable intermediate step

$$\int \frac{dy}{1+y^2} = \int 2t\cos(t^2)\,dt + C$$

$$y' = \underbrace{2t\cos(t^2)}_{\text{factor in } t}\,\underbrace{(1+y^2)}_{\text{factor in } y}$$
$$\frac{dy}{1+y^2} = 2t\cos(t^2)\,dt, \quad u = t^2$$
$$\arctan(y) = \sin(t^2) + C$$
$$y(\sqrt{\pi}) = 0 \implies 0 = \sin(\pi) + C \implies C = 0$$

Show a fully integrated answer

$$\arctan(y) = \sin(t^2)$$

3. ☐ Linear ☑ Separable $\quad \ln(y)y' = te^{t^2} \qquad y(1) = e$

Show a suitable intermediate step

$$\int \ln(y)\,dy = \int te^{t^2}\,dt + C$$

$$y' = \underbrace{te^{t^2}}_{\text{factor in } t}\ \underbrace{\frac{1}{\ln(y)}}_{\text{factor in } y}$$

$$\ln(y)\,dy = te^{t^2}\,dt, \quad u = t^2$$

$$\int \ln(y)\,dy = y\ln(y) - y + C$$

$$y\ln(y) - y = e^{t^2}/2 + C$$

$$y(1) = e \implies e\ln(e) - e = e/2 + C \implies C = -e/2$$

Show a fully integrated answer

$$y\ln(y) - y = (e^{t^2} - e)/2$$

4. ☑ Linear ☐ Separable $\quad ty' + (t+1)y = 1 \qquad y(1) = 1$

Show a suitable intermediate step

$$(te^t y)' = e^t$$

Standard form $y' + (1 + 1/t)y = 1/t$

$$y' + p(t)y = q(t), \quad p(t) = 1 + 1/t, \quad q(t) = 1/t$$

$$\mu(t) = \exp\left(\int p(t)\,dt\right)$$

$$\int (1 + 1/t)\,dt = t + \ln|t|$$

$$\mu(t) = e^{t+\ln|t|} = e^t|t| = te^t \quad (t > 0 \text{ since } t_0 = 1)$$

$$\int e^t\,dt = e^t + C$$

$$te^t y = e^t + C$$

$$y(1) = 1 \implies e = e + C \implies C = 0$$

Show a fully integrated answer

$$te^t y = e^t$$